A PROJECT REPORT ON

"Real-Time Wireless Audio Transmission Using ESP32"



B. Tech E.C.E. 2023-27

(VIth Sem)

Submitted by: GROUP - III

Shivam Dangwal (23001008062)

Om Gupta (23001008042)

Shubham Yadav (23001008074)

Nidhi (23001008038)

Riya (23001008053)

ABSTRACT

This project presents the design and implementation of a real-time wireless audio communication system using ESP32 microcontrollers, digital I2S audio interfaces, and the ESP-NOW communication protocol. The objective of the project is to develop a compact, low-latency, and efficient wireless voice transmission system that functions similarly to a digital walkie-talkie without requiring a conventional Wi-Fi network or internet connectivity.

The transmitter section of the system consists of an ESP32 microcontroller interfaced with an INMP441 I2S MEMS microphone. Audio signals are captured digitally at a sampling rate of 16 kHz using the I2S communication protocol, which ensures noise-resistant and high-quality audio acquisition. Since raw audio data requires high bandwidth for transmission, the system employs μ-law companding compression to convert 16-bit PCM audio samples into 8-bit compressed data. This significantly reduces the transmission payload while maintaining acceptable speech quality for voice communication applications.

The compressed audio packets are transmitted wirelessly using the ESP-NOW protocol, a lightweight peer-to-peer communication technology developed for ESP32 devices. ESP-NOW enables direct communication between devices with very low latency and without the need for routers or external infrastructure, making it highly suitable for real-time embedded communication systems.

On the receiver side, another ESP32 module receives the audio packets and performs μ-law decoding to reconstruct the original PCM audio samples. The decoded audio data is then transmitted through the I2S interface to a MAX98357A I2S DAC amplifier module connected to a speaker for real-time audio playback. To improve communication reliability, the system incorporates sequence numbering and simple packet loss concealment techniques. In case of packet loss or transmission errors, the receiver temporarily repeats the last valid audio frame to reduce audible glitches and maintain continuity of sound.

The project demonstrates the integration of embedded systems, wireless communication, digital signal processing, and real-time audio streaming into a single functional application. The implemented system achieves low-latency voice transmission with efficient bandwidth utilization and stable wireless communication. This project can be further enhanced with advanced audio codecs, noise suppression, adaptive buffering, and full-duplex communication to support more sophisticated wireless voice communication applications.

TABLE OF CONTENTS

Introduction

Objectives

System Architecture

Hardware Components

Software Implementation

Circuit Diagram

Working Principle

Advantages

Limitations

Applications

Future Scope

Conclusion

1. INTRODUCTION

Wireless communication systems have become an essential part of modern technology due to their wide range of applications in real-time data transfer, remote monitoring, industrial automation, and portable communication devices. Among these applications, real-time wireless voice communication systems such as walkie-talkies and intercom devices play a significant role in enabling fast and reliable communication without depending on complex network infrastructure. Traditional communication systems often require dedicated hardware, cellular networks, or internet connectivity, which may increase cost, complexity, and power consumption. Therefore, there is a growing need for low-cost, compact, and efficient wireless communication systems capable of transmitting audio data with minimal latency.

This project focuses on the design and implementation of a real-time wireless audio transmission system using ESP32 microcontrollers and digital I2S audio devices. The system utilizes an INMP441 I2S MEMS microphone for capturing audio signals and a MAX98357A I2S DAC amplifier module for audio playback through a speaker. Communication between the transmitter and receiver is established using the ESP-NOW protocol, which enables direct peer-to-peer wireless communication between ESP32 devices without requiring a Wi-Fi router or internet connection.

The primary objective of this project is to achieve efficient and low-latency voice transmission suitable for real-time communication applications. To reduce the bandwidth required for wireless transmission, the captured audio samples are compressed using μ-law companding techniques before transmission. On the receiver side, the compressed audio is decoded back into PCM audio samples and played through the speaker using the I2S audio interface. The system also implements packet sequencing and packet loss concealment methods to improve audio continuity and reliability during wireless communication.

The project combines concepts from embedded systems, wireless networking, digital signal processing, and real-time audio streaming into a single integrated application. By using digital audio interfaces and lightweight wireless protocols, the proposed system provides better noise immunity, efficient bandwidth utilization, and reduced communication delay compared to conventional analog communication methods.

This work demonstrates how low-cost microcontrollers such as the ESP32 can be used to develop practical wireless voice communication systems with real-time performance. The implemented system can serve as a foundation for future developments in wireless intercom systems, IoT-based communication devices, smart surveillance systems, and portable embedded audio applications.

2. OBJECTIVES

The specific objectives of the project are as follows:

1. To capture real-time audio signals using an INMP441 I2S MEMS microphone interfaced with the ESP32 microcontroller.
2. To implement digital audio acquisition and transmission using the I2S (Inter-IC Sound) communication protocol for improved audio quality and reduced noise interference.
3. To establish wireless communication between two ESP32 devices using the ESP-NOW protocol for direct peer-to-peer data transfer.
4. To reduce transmission bandwidth requirements by implementing μ-law audio compression techniques for efficient audio data encoding.
5. To decode the compressed audio data at the receiver side and reproduce the audio signal using a MAX98357A I2S DAC amplifier and speaker system.
6. To achieve low-latency real-time voice communication suitable for walkie-talkie and intercom applications.
7. To implement packet sequencing and packet loss concealment methods in order to improve communication reliability and maintain audio continuity during wireless transmission.
8. To study and integrate concepts related to embedded systems, wireless networking, digital signal processing, and real-time audio streaming into a practical hardware application.
9. To develop a scalable foundation for future improvements such as full-duplex communication, noise suppression, advanced audio codecs, and IoT-based wireless communication systems.
10. To demonstrate the practical implementation of a low-cost wireless voice communication system using commercially available embedded hardware modules.

3. SYSTEM ARCHITECTURE

The proposed wireless audio communication system is designed using two ESP32 microcontrollers functioning as the transmitter and receiver units. The overall architecture integrates digital audio acquisition, wireless data transmission, audio compression, and real-time playback into a unified embedded communication system. The system is divided into two major sections: the transmitter section and the receiver section.

1. Transmitter Section

The transmitter unit is responsible for capturing and transmitting audio data wirelessly. It consists of the following main components:

a) INMP441 I2S MEMS Microphone

The INMP441 digital microphone is used for capturing voice signals. Unlike analog microphones, the INMP441 provides digital audio output using the I2S communication protocol, which reduces noise interference and improves audio quality. The microphone samples the incoming audio signal at a frequency of 16 kHz.

b) ESP32 Microcontroller

The ESP32 acts as the central processing unit of the transmitter. It performs the following operations:

- Receives digital audio data from the microphone through the I2S interface.
- Processes the audio samples in real time.
- Applies μ-law companding compression to reduce the size of audio data.
- Creates data packets containing sequence numbers and compressed audio samples.
- Transmits the packets wirelessly using the ESP-NOW protocol.

c) ESP-NOW Wireless Communication

ESP-NOW is used as the wireless communication protocol between the transmitter and receiver ESP32 modules. It enables direct peer-to-peer communication without requiring a Wi-Fi router or internet connection. This protocol provides low-latency communication suitable for real-time voice transmission.

---

2. Receiver Section

The receiver unit is responsible for receiving, decoding, and reproducing the transmitted audio signals.

a) ESP32 Receiver Module

The receiver ESP32 receives the audio packets transmitted through ESP-NOW. It performs the following functions:

- Detects incoming packets and verifies packet sequence numbers.
- Identifies packet loss during transmission.
- Applies packet loss concealment by replaying the previous audio frame when packets are lost.
- Decodes the received μ-law compressed audio back into 16-bit PCM audio samples.
- Sends the decoded audio samples to the DAC amplifier through the I2S interface.

b) MAX98357A I2S DAC Amplifier

The MAX98357A module acts as both a Digital-to-Analog Converter (DAC) and an audio amplifier. It converts the digital PCM audio received from the ESP32 into analog audio signals and amplifies them to drive the speaker.

c) Speaker Unit

The amplified analog audio signal is finally reproduced through a speaker, enabling real-time voice playback at the receiver side.

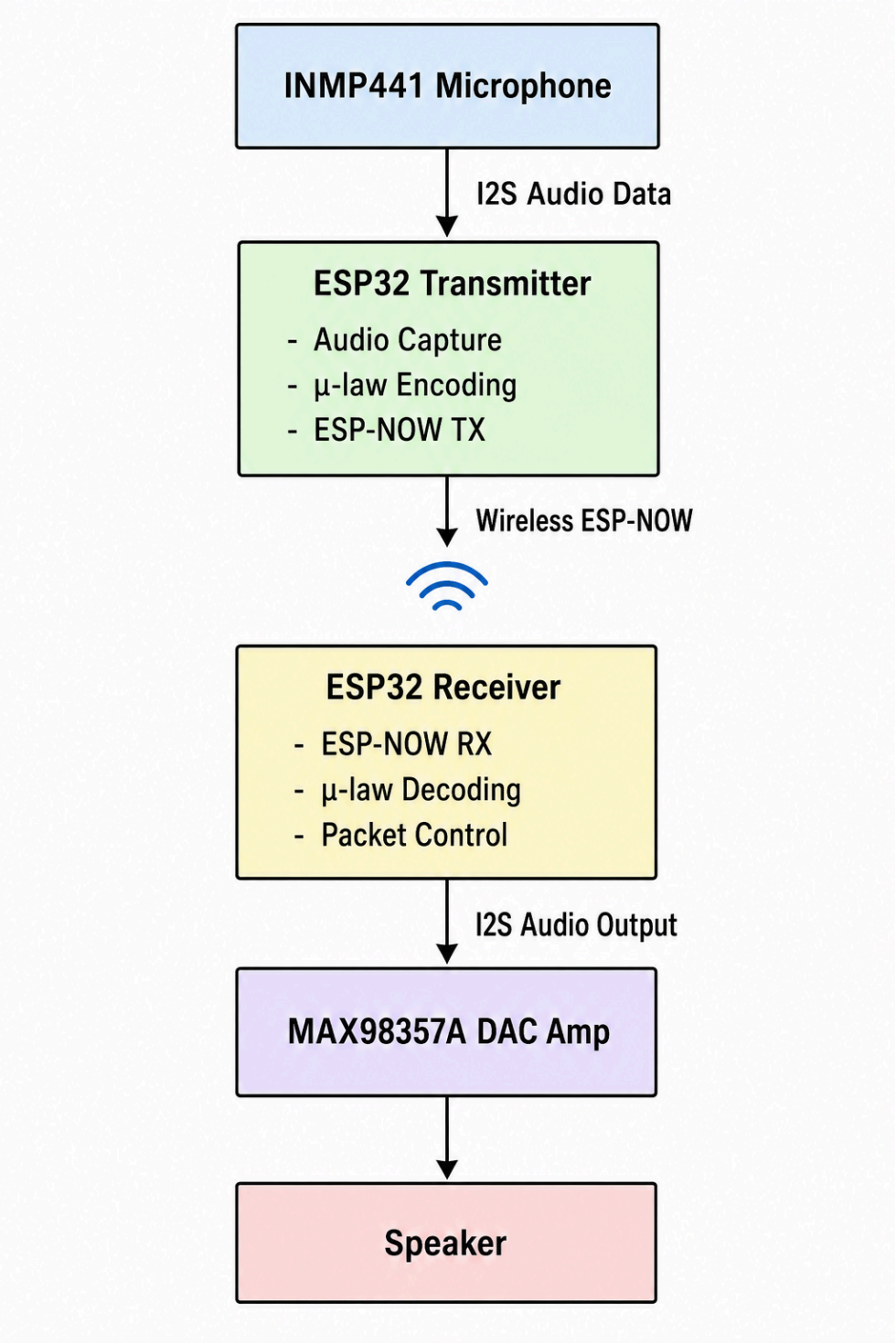
Working Principle of the System

The microphone captures voice signals and converts them into digital audio samples using the I2S interface. The ESP32 transmitter compresses these samples using μ-law encoding and sends them wirelessly using ESP-NOW packets. Each packet contains a sequence number and compressed audio data for efficient transmission.

At the receiver side, the ESP32 receives the packets and checks the sequence numbers to detect any missing packets. The compressed audio is then decoded back into PCM format and transmitted to the MAX98357A DAC amplifier through the I2S interface.

The amplifier converts the digital data into analog signals and drives the speaker for audio playback.

Block Diagram of the System



The proposed architecture provides a compact and efficient solution for real-time wireless voice communication while maintaining low latency, reduced bandwidth usage, and reliable audio playback.

5. HARDWARE COMPONENTS

1. ESP32 Microcontroller

The ESP32 is the primary controller used in both the transmitter and receiver sections of the system. It is a low-cost and high-performance microcontroller developed by with built-in Wi-Fi and Bluetooth capabilities.

Features of ESP32

- Dual-core processor
- Integrated Wi-Fi and Bluetooth
- Support for ESP-NOW communication
- Multiple GPIO pins
- Built-in I2S peripheral support
- Low power consumption
- High processing speed suitable for real-time applications

Role in the Project

In the transmitter unit, the ESP32 captures audio from the microphone, compresses it using μ-law encoding, and transmits it wirelessly using ESP-NOW. In the receiver unit, the ESP32 receives the audio packets, decodes them, and sends the audio data to the DAC amplifier for playback.

---

2. INMP441 I2S MEMS Microphone

The INMP441 is a digital MEMS microphone module with an integrated I2S interface. It converts sound waves into digital audio signals that can be directly processed by the ESP32.

Features of INMP441

- Digital I2S output
- Low noise performance
- Compact size
- Low power operation
- High sensitivity
- Suitable for voice recording applications

Role in the Project

The microphone captures voice signals from the user and sends digital audio samples to the ESP32 transmitter through the I2S interface. Since the output is digital, it provides better noise immunity compared to analog microphones.

---

3. MAX98357A I2S DAC Amplifier

The MAX98357A is a digital-to-analog converter (DAC) with an integrated Class-D audio amplifier. It receives digital audio data through I2S and converts it into amplified analog audio signals.

Features of MAX98357A

- I2S digital audio input
- Built-in DAC and amplifier
- High efficiency Class-D amplification
- Compact and low-power design
- Direct speaker driving capability

Role in the Project

The MAX98357A receives PCM audio data from the ESP32 receiver and converts it into analog audio signals. The built-in amplifier increases the signal strength to drive the speaker for real-time audio playback.

---

4. Speaker

A speaker is used at the receiver side to reproduce the received audio signals.

Features

- Converts electrical audio signals into sound waves
- Compatible with low-power audio amplifiers
- Suitable for speech playback

Role in the Project

The speaker receives amplified analog signals from the MAX98357A module and produces audible sound output for the user.

Summary of Hardware Components

Component	Function
ESP32	Main controller and wireless communication
INMP441 Microphone	Captures digital audio input
MAX98357A DAC Amplifier	Converts and amplifies digital audio
Speaker	Produces audio output

The integration of these hardware components enables the successful implementation of a compact and efficient wireless audio communication system capable of low-latency real-time voice transmission.

6. SOFTWARE IMPLEMENTATION

1. Development Environment

The software for the project was developed using the following tools and frameworks:

Software Tool	Purpose
Arduino IDE	Program development and uploading
ESP32 Board Package	ESP32 support libraries
Embedded C/C++	Programming language
WiFi Library	Wi-Fi configuration
ESP-NOW Library	Wireless communication
I2S Driver Library	Digital audio interfacing

The ESP32 board package was installed in the Arduino IDE to enable programming and configuration of ESP32 microcontrollers.

---

2. Transmitter Software Implementation

The transmitter software captures audio from the INMP441 I2S microphone, compresses the audio data, and transmits it wirelessly to the receiver.

a) I2S Audio Acquisition

The I2S peripheral of the ESP32 is configured in receive mode to capture digital audio samples from the microphone at a sampling frequency of 16 kHz. The microphone provides 32-bit audio samples which are read into a buffer using DMA-supported I2S communication.

The software continuously reads blocks of audio samples from the microphone for real-time processing.

b) Audio Compression Using μ-law Encoding

Since raw PCM audio requires large bandwidth for wireless transmission, μ-law companding compression is implemented in software. The captured 16-bit audio samples are compressed into 8-bit encoded samples before transmission.

This compression technique significantly reduces the packet size while maintaining acceptable speech quality for voice communication applications.

c) Packet Formation

The software creates fixed-size data packets for wireless transmission. Each packet contains:

- A sequence number
- Compressed audio samples

The sequence number helps the receiver identify packet loss and maintain audio continuity.

d) ESP-NOW Transmission

The compressed audio packets are transmitted using the ESP-NOW protocol. The transmitter sends packets directly to the receiver ESP32 using its MAC address. ESP-NOW enables low-latency peer-to-peer wireless communication without requiring a router or internet connectivity.

---

3. Receiver Software Implementation

The receiver software receives wireless audio packets, reconstructs the audio signal, and outputs the audio through the speaker.

a) ESP-NOW Packet Reception

The receiver ESP32 continuously listens for incoming ESP-NOW packets. A callback-based reception mechanism is implemented to process incoming packets immediately after reception.

b) Packet Loss Detection

Each packet contains a sequence number that is checked at the receiver side. If sequence numbers are not consecutive, the software identifies packet loss during wireless communication.

c) Packet Loss Concealment

To minimize audible interruptions caused by missing packets, a simple packet loss concealment mechanism is implemented. In case of packet loss, the receiver temporarily repeats the previous audio frame to maintain smooth audio playback.

d) μ-law Audio Decoding

The received compressed audio samples are decoded back into 16-bit PCM audio data using μ-law decoding algorithms. This restores the original audio waveform for playback.

e) I2S Audio Output

After decoding, the PCM audio samples are transmitted through the I2S interface to the MAX98357A DAC amplifier module. The amplifier converts the digital audio into analog signals and drives the speaker.

---

4. Real-Time Processing

The entire system is designed for real-time operation. The software uses:

- DMA-supported I2S communication
- Callback-based packet reception
- Fixed-size packet streaming
- Efficient memory management

These techniques reduce processing overhead and help maintain low audio latency during communication.

---

5. Software Workflow

The overall software workflow of the system is as follows:

Microphone Input
↓
I2S Audio Sampling
↓
μ-law Compression
↓
Packet Formation
↓
ESP-NOW Transmission
↓
Wireless Reception
↓
Packet Verification
↓
μ-law Decoding
↓
I2S Audio Output
↓
Speaker Playback

---

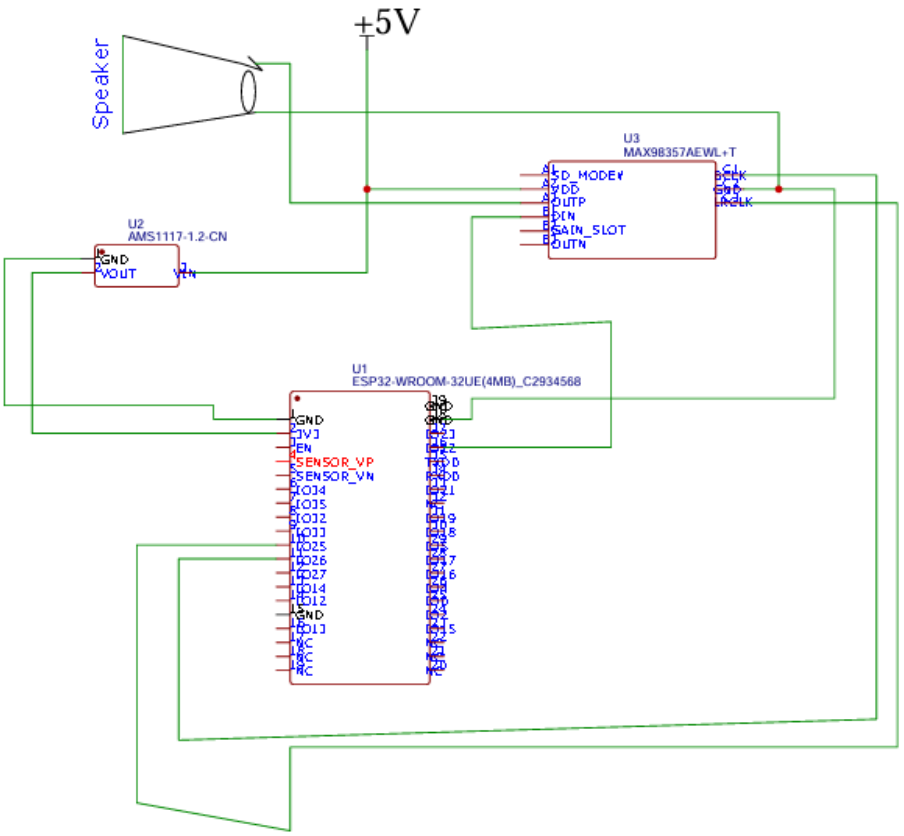
6. Advantages of the Software Design

The implemented software architecture provides several advantages:

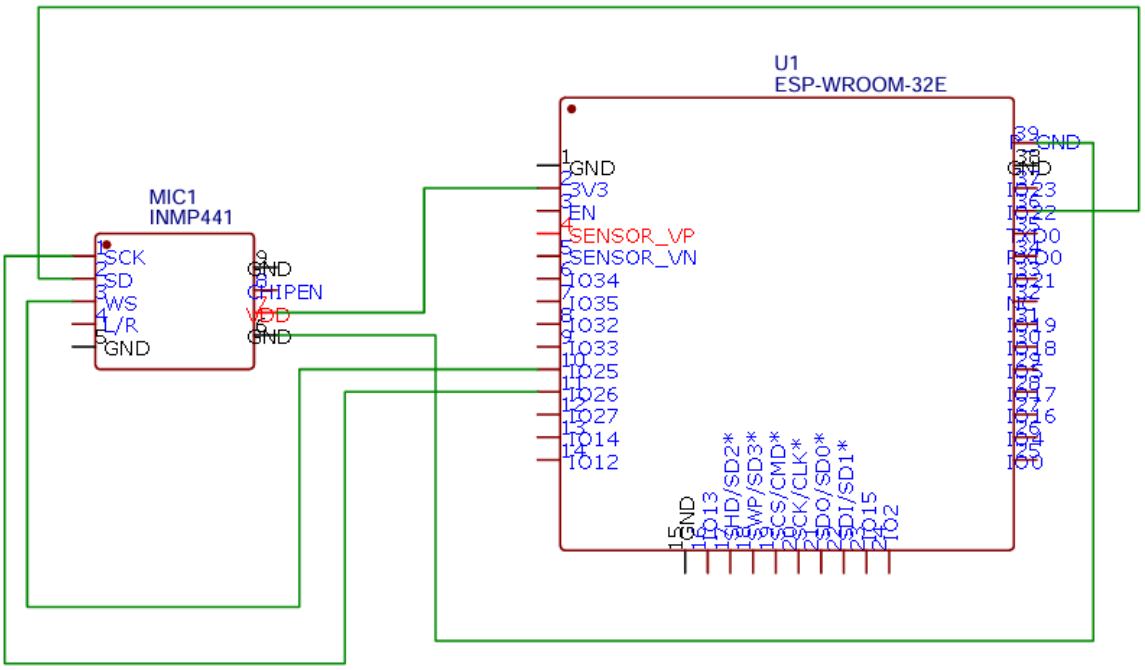
- Low-latency wireless communication
- Efficient bandwidth utilization
- Reduced packet size through compression
- Real-time audio streaming capability
- Simple and lightweight implementation
- Reliable packet handling using sequence numbers
- Improved audio continuity through packet loss concealment

The software implementation successfully integrates digital audio processing and wireless embedded communication into a compact and efficient real-time voice transmission system.

7. CIRCUIT DIAGRAM



RECEIVER SIDE



TRANSMITTER SIDE

8. WORKING PRINCIPLE

The working principle of the proposed wireless audio communication system is based on real-time digital audio acquisition, wireless transmission using ESP-NOW, audio compression using μ-law encoding, and audio playback through an I2S DAC amplifier. The system operates by continuously capturing voice signals at the transmitter side, transmitting the processed audio wirelessly, and reproducing the audio at the receiver side with minimal delay.

The complete operation of the system can be divided into the following stages:

---

1. Audio Signal Acquisition

The process begins at the transmitter side where the INMP441 I2S MEMS microphone captures the user's voice signals. The microphone converts sound waves into digital audio samples using the I2S communication protocol.

Unlike analog microphones, the INMP441 directly produces digital audio data, which improves noise immunity and eliminates the need for external analog-to-digital conversion circuitry. The microphone samples the audio signal at a rate of 16 kHz, which is suitable for speech communication applications.

The ESP32 microcontroller receives these digital audio samples through its built-in I2S peripheral configured in receive mode.

---

2. Audio Processing and Compression

The captured audio samples are initially available as 32-bit digital PCM samples. Since transmitting raw audio data requires high bandwidth, the ESP32 performs audio compression using the μ-law companding algorithm.

In this process:

- 16-bit linear PCM audio samples are compressed into 8-bit encoded values.
- The compression reduces packet size and wireless bandwidth usage.

- The logarithmic nature of μ-law encoding preserves speech quality while reducing data size.

The compressed audio data allows efficient real-time wireless transmission over ESP-NOW.

---

3. Packet Formation

After compression, the ESP32 organizes the audio data into fixed-size packets. Each packet contains:

- A sequence number
- Compressed audio samples

The sequence number is incremented for every transmitted packet and helps the receiver detect packet loss or missing audio frames during communication.

The use of fixed-size packets simplifies synchronization and maintains continuous audio streaming.

---

4. Wireless Transmission Using ESP-NOW

The compressed audio packets are transmitted wirelessly using the ESP-NOW protocol. ESP-NOW enables direct communication between ESP32 devices without requiring a Wi-Fi router or internet connection.

The transmitter sends packets directly to the MAC address of the receiver ESP32. Since ESP-NOW operates with low communication overhead and minimal latency, it is suitable for real-time voice transmission applications such as walkie-talkies and wireless intercom systems.

---

5. Packet Reception and Verification

At the receiver side, the ESP32 continuously listens for incoming ESP-NOW packets. When a packet is received, the receiver extracts the sequence number and compressed audio payload.

The sequence numbers are checked to determine whether any packets were lost during wireless transmission. If the receiver detects missing packets, it identifies a packet loss condition.

---

6. Packet Loss Concealment

Wireless communication may occasionally result in lost or delayed packets due to interference or transmission errors. To minimize audible distortions caused by missing packets, the system implements a simple packet loss concealment technique.

If a packet is lost:

- The receiver temporarily reuses the previously received audio frame.
- This reduces clicks, gaps, and interruptions in the audio output.
- The method improves continuity of real-time speech playback.

---

7. Audio Decoding

The received compressed audio samples are decoded using the μ-law decoding algorithm. The decoder reconstructs the original 16-bit PCM audio samples from the 8-bit compressed data.

The decoded samples are then prepared for audio playback.

---

8. Audio Playback

The ESP32 receiver sends the decoded PCM audio samples through the I2S interface to the MAX98357A DAC amplifier module.

The MAX98357A performs two operations:

- Converts digital PCM audio into analog audio signals
- Amplifies the audio signal to drive the speaker

Finally, the speaker reproduces the received voice signal in real time.

Overall Operational Flow

Voice Input
↓
INMP441 Microphone
↓
I2S Digital Audio Capture
↓
ESP32 Transmitter
↓
μ-law Audio Compression
↓
Packet Formation
↓
ESP-NOW Wireless Transmission
↓
ESP32 Receiver
↓
Packet Verification & Loss Handling
↓
μ-law Audio Decoding
↓
I2S Audio Output
↓
MAX98357A DAC Amplifier
↓
Speaker Output

The implemented working principle enables low-latency, efficient, and reliable wireless audio communication using compact embedded hardware modules and lightweight wireless protocols.

10. ADVANTAGES

1. Low-Latency Communication

The system uses the ESP-NOW protocol, which provides direct peer-to-peer wireless communication between ESP32 devices. Since the communication does not require a router or internet connection, the transmission delay is significantly reduced. This enables real-time voice communication suitable for walkie-talkie and intercom applications.

---

2. Infrastructure-Free Communication

One of the major advantages of the system is that it operates independently of external network infrastructure. Unlike traditional wireless communication systems, the proposed system does not require:

- Wi-Fi routers
- Cellular networks
- Internet connectivity

This makes the system useful in remote locations and temporary communication setups.

---

3. Low-Cost Implementation

The system is developed using low-cost and widely available hardware components such as:

- ESP32 microcontrollers
- INMP441 microphone module
- MAX98357A DAC amplifier

The use of affordable embedded hardware reduces the overall system cost while maintaining good performance.

---

4. Digital Audio Communication

The use of I2S digital audio interfaces improves audio quality and reduces noise interference compared to analog communication systems. Since the microphone output is digital, signal degradation due to analog noise is minimized.

---

5. Efficient Bandwidth Utilization

The implementation of μ-law audio compression reduces the size of transmitted audio data. This:

- Reduces wireless bandwidth requirements
- Improves transmission efficiency
- Enables smoother real-time audio streaming

The compression technique helps maintain acceptable speech quality while minimizing packet size.

---

6. Compact and Portable Design

The hardware modules used in the system are compact and lightweight. The overall design can be easily integrated into portable communication devices powered by batteries or USB power supplies.

---

7. Real-Time Audio Streaming

The system continuously captures, processes, transmits, and reproduces audio with minimal buffering. The use of DMA-supported I2S communication and callback-based packet handling enables efficient real-time audio streaming.

---

8. Packet Loss Handling

The receiver implements packet sequence verification and simple packet loss concealment techniques. In case of missing packets, the previous audio frame is replayed temporarily to reduce interruptions in audio playback.

11. LIMITATIONS

1. Limited Communication Range

ESP-NOW communication range is limited compared to long-range wireless technologies. Obstacles such as walls, interference, and environmental conditions can reduce signal strength and communication distance.

---

2. No Internet Connectivity

Although infrastructure-free communication is an advantage, the system cannot communicate over long distances through the internet. The communication is limited to the wireless range supported by ESP-NOW.

---

3. Basic Audio Compression

The system uses μ-law compression, which is simpler than modern audio codecs such as Opus or AAC. While suitable for speech communication, the audio quality is lower compared to advanced compression algorithms.

---

4. Mono Audio Transmission

The implemented system supports only mono audio transmission. Stereo audio communication is not implemented in the current design.

---

5. Packet Loss During Transmission

Wireless communication is susceptible to packet loss due to:

- Signal interference
- Weak wireless signals
- Environmental disturbances

Although packet concealment is implemented, excessive packet loss may still affect audio quality.

---

6. No Advanced Noise Cancellation

The system does not include advanced audio processing features such as:

- Noise suppression
- Echo cancellation
- Automatic gain control

As a result, environmental noise may affect communication clarity.

---

7. Limited Scalability

The current implementation is primarily designed for communication between two ESP32 devices. Expanding the system to support multiple simultaneous users would require additional synchronization and network management mechanisms.

---

8. Power Consumption

Continuous wireless communication and real-time audio processing increase power consumption, especially when battery-powered operation is used for long durations.

---

13. FUTURE SCOPE

The proposed wireless audio communication system successfully demonstrates real-time voice transmission using ESP32 microcontrollers, I2S digital audio interfaces, and the ESP-NOW communication protocol. Although the current implementation provides reliable short-range audio communication with low latency, there are several opportunities for further improvement and expansion. Future enhancements can significantly improve audio quality, communication range, system reliability, and overall functionality.

---

1. Implementation of Advanced Audio Codecs

The current system uses μ-law audio compression, which is simple and efficient for speech transmission. However, more advanced audio codecs such as:

- ADPCM
- Opus
- AAC
- SBC

can be implemented in the future to improve audio quality while maintaining low bandwidth usage. Advanced codecs can provide clearer voice transmission and better noise handling.

---

2. Full-Duplex Communication

The existing system mainly supports simplex or one-way communication at a time. Future development can enable full-duplex communication, allowing simultaneous transmission and reception of audio similar to mobile phone communication systems.

This enhancement would require:

- Simultaneous audio capture and playback
- Better synchronization
- Efficient buffer management

---

3. Noise Suppression and Echo Cancellation

Environmental noise and acoustic feedback can affect communication clarity. Future versions of the system can include digital signal processing techniques such as:

- Noise suppression
- Echo cancellation
- Automatic gain control (AGC)
- Voice enhancement filters

These techniques would improve speech intelligibility and overall audio quality.

---

4. Increased Communication Range

The communication range of ESP-NOW is limited by Wi-Fi signal strength and environmental conditions. Future improvements may include:

- External high-gain antennas
- Mesh networking
- Wi-Fi repeater integration
- Long-range wireless modules

These enhancements can increase operational distance and improve communication reliability.

---

5. Multi-User Communication System

The present implementation mainly supports communication between two ESP32 devices. Future work can extend the system to support:

- Multiple users
- Group communication
- Broadcast communication
- Conference-style audio transmission

This would require improved network management and packet routing mechanisms.

---

6. Battery-Powered Portable Design

The current prototype can be further developed into a compact portable device powered by rechargeable batteries. Future versions may include:

- Lithium-ion battery support
- Battery charging circuits
- Power management systems
- Low-power operating modes

This would make the system more suitable for portable field communication applications.

---

7. Mobile Application Integration

Future systems can be integrated with Android or iOS applications using Wi-Fi or Bluetooth connectivity. A mobile application could provide:

- Device monitoring
- Audio recording
- Configuration settings
- Push-to-talk controls
- Real-time communication management

This would improve user interaction and system usability.

---

8. Internet-Based Communication

The current design operates without internet connectivity. Future enhancements can integrate:

- Wi-Fi networking
- MQTT protocols
- VoIP communication
- Cloud connectivity

This would allow long-distance communication over the internet and IoT-based voice communication applications.

---

9. Secure Audio Transmission

At present, the transmitted audio packets are not encrypted. Future implementations can add:

- Data encryption
- Secure authentication
- Secure peer pairing
- Protected wireless communication

These security features would improve privacy and prevent unauthorized access.

---

10. AI-Based Audio Processing

Artificial intelligence and machine learning techniques can be integrated in future versions for:

- Voice recognition
- Speaker identification
- Automatic noise filtering
- Voice activity detection
- Smart audio optimization

These advanced features can significantly enhance system intelligence and performance.

---

11. Real-Time Audio Recording and Storage

Future systems may include memory card support or onboard storage for:

- Audio recording
- Playback
- Voice logging
- Emergency communication storage

This feature would be useful for surveillance, security, and communication analysis applications.

14. CONCLUSION

The wireless audio communication system developed in this project successfully demonstrates real-time voice transmission using ESP32 microcontrollers, I2S digital audio interfaces, and the ESP-NOW communication protocol. The project integrates embedded systems, wireless networking, and digital audio processing to create a compact and efficient communication platform capable of transmitting voice signals with low latency and acceptable audio quality.

The transmitter section effectively captures voice signals using the INMP441 I2S MEMS microphone and processes the audio data through μ-law compression techniques to reduce bandwidth requirements during wireless transmission. The ESP-NOW protocol enables direct peer-to-peer communication between ESP32 devices without requiring external network infrastructure such as routers or internet connectivity. On the receiver side, the transmitted audio packets are decoded and reproduced using the MAX98357A I2S DAC amplifier and speaker system.

The implementation of packet sequencing and packet loss concealment techniques improves communication reliability and helps maintain continuous audio playback even in the presence of wireless transmission errors. The use of digital audio communication through the I2S interface provides better noise immunity and stable audio performance compared to conventional analog systems.

The project achieves the primary objective of designing a low-cost, portable, and real-time wireless voice communication system suitable for applications such as walkie-talkies, wireless intercoms, portable communication devices, and IoT-based audio systems. The system also demonstrates efficient bandwidth utilization and practical real-time performance using lightweight embedded hardware and communication protocols.

Although the current implementation has certain limitations related to communication range, audio compression quality, and advanced audio processing features, it provides a strong foundation for future improvements. Additional enhancements such as advanced codecs, noise suppression, secure communication, full-duplex transmission, and internet integration can further expand the capabilities of the system.

Overall, the project successfully validates the feasibility of implementing a reliable and efficient wireless audio streaming system using low-cost embedded hardware. It also provides valuable practical knowledge in the fields of embedded systems,

wireless communication, digital signal processing, and real-time audio application development.